Chapter 56. Ventilator-Supported Speech

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Ventilator-Supported Speech: Introduction

Under normal circumstances, the respiratory system supplies the aeromechanical drive that allows the vocal folds, tongue, lips, and other structures to create the sounds of speech. Although a simplification, this drive can be understood as the tracheal pressure. Usually this pressure is exquisitely and actively controlled by muscles of the chest wall. When speech is produced with ventilator support, however, the ventilator and the respiratory system must work together to produce the pressure that drives speech production. In most cases, this pressure is markedly different from that of normal speech production. As a result, the act of speaking can be challenging for patients, and they often require assistance from their pulmonologists, respiratory therapists, and speech-language pathologists. The views presented here are that patients should be enabled to speak whenever possible, and that ventilator-supported speech can often be improved by using interventions such as those described in this chapter.

The Importance of Speech

Communication is critical to maintaining good quality of life. Communication is the key to being able to express needs, participate in social activities, and retain control over important life decisions.[1–5] It can be especially important in intensive care or during end-of-life care.[6,7] There are many ways to communicate, but of these, speech is the fastest and most convenient. It works in the dark and across telephone lines, and it conveys meaning and emotions through words and tone of voice (i.e., pitch, loudness, quality, and timing).

Patients who are ventilator-supported often complain of speech problems, especially unwanted pauses and inadequate loudness.[8] This is illustrated in the following quotation,[9] which comes from a patient who was asked, “Do you have any problems with your speech?” Note that the pauses between phrases were 3 to 4 seconds (• = 1 second), and that some words were produced without voice (shown in parentheses). To appreciate the severity of the speech problem, this quotation should be read aloud with timed pauses:

“Yes •••• Um, getting cut (off) •••• people interfering with me ••• trying to finish my sentences (for me) •••• that’s the most frustrating (part).”

Fortunately, it is nearly always possible to improve ventilator-supported speech. The following quotation comes from the same patient, after she received speech interventions. Pauses were reduced to approximately 1 second, and the amount of speech per breath increased by approximately 50%.

“Since I’ve been able to talk better • I’ve par- participated in talking with more people and • and in conversation instead of just doing the listening • and talking as little as possible • I’ve joined in and, and • I don’t know, I’ve just been part of a group more • Very few people try to finish my sentences now and • second guess what I’m going to say • It’s, it’s helped a lot.”

Fundamentals of Speech Breathing

Speech comprises the tones, hisses, pops, and buzzes that are the acoustic representation of language.[10] Speech is usually produced by the coordinated efforts of more than 100 muscles spanning the chest wall, larynx, and upper airway structures (pharynx, velum, jaw, lips, and tongue). Speech breathing is the process by which the driving forces are supplied to downstream structures to generate speech, and it is tailored to simultaneously serve speech-related functions and meet gas exchange requirements.[11]

The normal speech breathing cycle has an inspiratory phase ranging from 0.5 to 0.7 second.[12–16] Inspirations during speaking are so short that the pauses they create are hardly noticeable. The longer, more variable expiratory phase of the cycle is the speaking phase. Its duration averages 3 to 5 seconds, but varies substantially as a function of linguistic factors, cognitive variables, mechanical constraints, and ventilatory needs.[13,14,16–18] Speech is usually produced throughout most of the expiratory phase, although nonspeech expirations are likely to occur in senescent subjects,[19] with high cognitive-linguistic demands,[20] and under conditions of elevated ventilatory drive.[16,21,22] The tidal volumes used in speech breathing are typically about twice as large as those of resting tidal breathing.[23]

The normal speech breathing cycle has a characteristic tracheal pressure profile. During inspiration, pressure is substantially and briefly negative, and during expiration (speaking), it is moderately positive and relatively steady. For speech of conversational loudness, pressure is generally in the range of 5 to 10 cm H_2O.[24,25] Pressure usually remains relatively constant throughout expiration so that loudness and voice quality remain relatively constant.[26,27] An increase in pressure generally causes an increase in loudness. To produce voiced speech sounds (such as vowels), the vocal folds must oscillate, which requires a minimum pressure of approximately 2 cm H_2O.[28,29] When voiced sounds are produced with a constricted oral airway (such as the plosive /d/ or the fricative /v/), slightly higher pressures are needed.

The pressures of normal speech breathing are controlled actively throughout the cycle by muscles of the chest wall.[27] Inspiration is driven by the diaphragm (with activation of other muscles that serve to stabilize the chest wall), and expiration (speaking) is driven by both expiratory rib cage and abdominal muscular pressures (with the latter predominating) when in upright body positions, and by expiratory rib cage muscular pressure alone when in the supine body position. These muscular pressures are usually supplemented by positive recoil pressure, because most conversational speech is produced above the resting expiratory level. On rare occasions, when speech is initiated at a very large lung volume (i.e., near total lung capacity), positive recoil force may be so great that inspiratory muscular pressure must be called into play to counteract (i.e., “brake”) the high positive pressure.

A Framework for Understanding Ventilator-Supported Speech

During ventilator-supported speech breathing, tracheal pressure reflects the combined contributions of the patient’s prevailing respiratory recoil pressure, active muscular pressures (if the patient is able to activate chest wall muscles), and inspiratory and/or expiratory drive from the ventilator, among other factors. A useful framework for understanding, evaluating, and managing ventilator-supported speech is to relate speech to tracheal pressure.[9,11,30]

Speech with Invasive Positive-Pressure Ventilation

Invasive positive-pressure ventilation (InPPV) is used here to mean intermittent positive-pressure ventilation via tracheostomy. Because an endotracheal tube prohibits vocalizing and can damage the vocal folds, speech is one of several factors to consider when deciding whether and when to perform a tracheotomy (see Chapter 40). Speaking with InPPV is substantially different from speaking with other forms of ventilation, but it can be very successful. The primary differences stem from the fact that the ventilator-delivered air enters below the larynx.

Deflate the Tracheostomy Tube Cuff

For a patient to speak, the first and most critical step is to configure the tracheostomy tube so that flow (and thus pressure) can reach the larynx. This means that the cuff on the tube must be deflated (and/or the tube must contain a fenestration). If cuff deflation or use of a fenestrated tube is deemed inadvisable, alternative forms of speech or communication must be sought (see Chapter 40 and “Alternatives to Ventilator-Supported Speech”).

Cuff deflation is feasible in most chronically ventilated patients[31,32] and in acutely ventilated patients who are otherwise stable. Indeed, in some patients, the cuff can be deflated and speech training initiated on the first day after the tracheotomy has been performed. Unfortunately, cuff deflation and speech training are often delayed (or not performed at all) because of unfamiliarity with the procedure, concerns about aspiration, and fear of causing hypoventilation and hypoxemia. Hypoventilation and hypoxemia can be avoided by increasing the ventilator-delivered tidal volume. In many stable patients, the cuff can remain deflated all day (and inflated only for sleep). In less-stable patients, cuff deflation may need to be brief (to allow communication with family members or caregivers) and performed only in the presence of a pulmonologist or respiratory therapist (see Clinical Scenario 1). It is sometimes helpful to deflate the cuff only partially in patients who tend to lose too much of the inspired tidal volume through an open larynx. The recommended procedure for initial cuff deflation is described in “Important Cautions”.

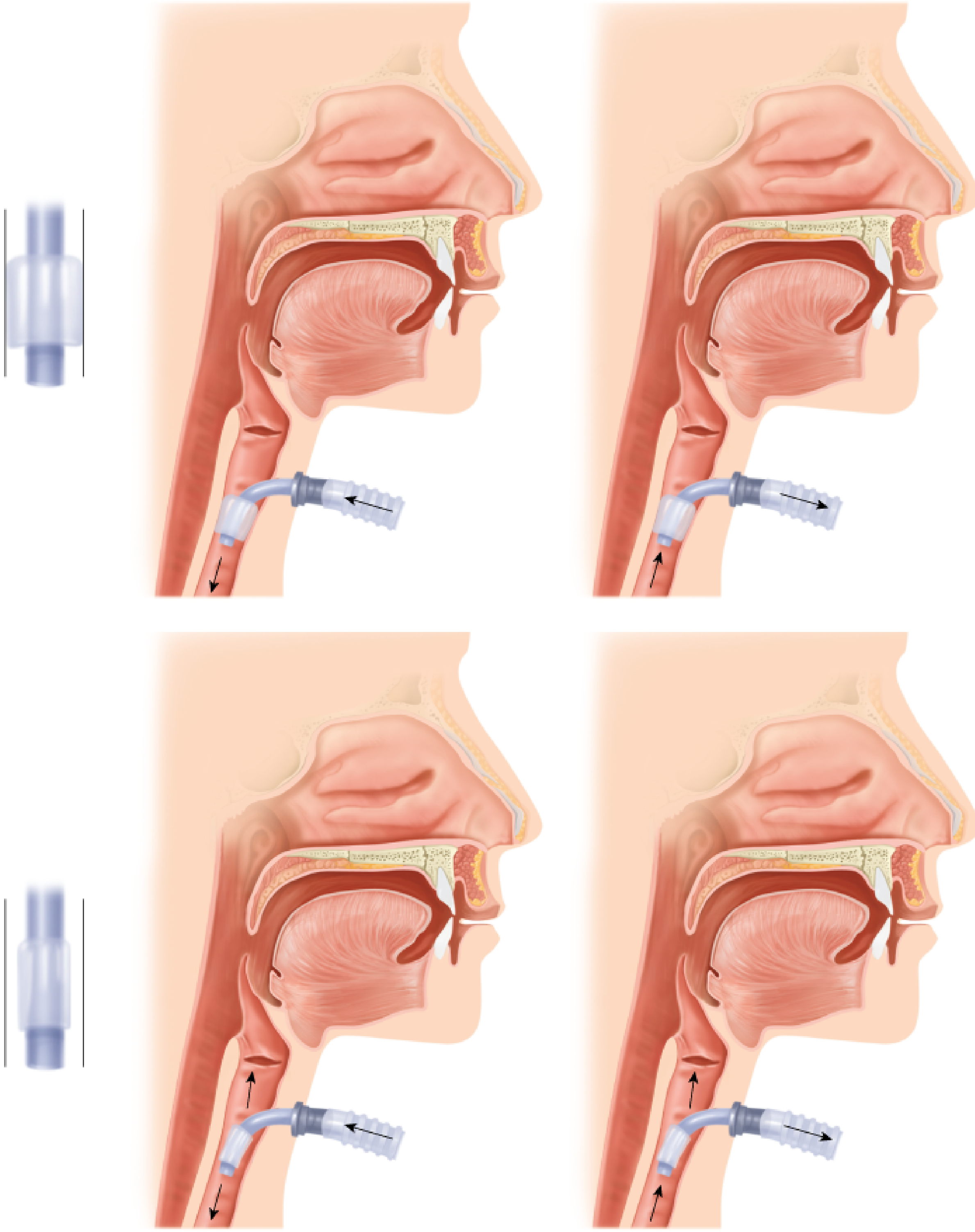
Clinical Scenario 1

> Mrs. Z was a 52-year-old woman with breast cancer that had metastasized to her lungs, causing her to be very short of breath and unable to breathe on her own. She obtained relief from assisted ventilation via a tracheostomy and could not be weaned from the ventilator. She was frustrated by not being able to speak to her caregivers, husband, and daughters about the nature and intensity of the care she wished to receive and about her impending death. Communication by writing was tedious. She was advised that the cuff of her tracheostomy tube could be deflated so that she could speak, but that she would likely experience periods of breathlessness at first while ventilator adjustments were being made. She was very fearful of this, but with encouragement and reassurance the procedure was successful. For several minutes at a time, with only slight shortness of breath, she was able to speak and communicate her wishes to her family. She went home on the ventilator and died peacefully there.

After cuff deflation, the patient’s voice should be assessed. If voice problems are noted, the resistance to “bypass” gas flow around the cuff should be estimated to determine if it is sufficiently low to allow good vocalizing. Bypass resistance can be estimated by blocking expiratory flow where it exits the ventilator and observing tracheal pressure during expiration. When the resistive pressure drop is less than 5 cm H_2O, cuff deflation is generally successful, but when pressure exceeds 10 cm H_2O, cuff deflation often fails. Bypass resistance can usually be reduced by downsizing the tracheostomy tube. If the patient still cannot vocalize, the upper airway, larynx, and upper trachea should be viewed via nasendoscopy to rule out conditions such as excessive secretions, laryngeal edema, granulation tissue, tracheal scar formation, or vocal fold paralysis.

Figure 56-1 illustrates the flow routes for inspiration and expiration with an inflated cuff (closed system) and deflated cuff (open system). When the cuff is inflated, no flow reaches the larynx. When the cuff is deflated, flow reaches the larynx during inspiration and expiration, making it theoretically possible to produce speech during both phases of the ventilator’s cycle. The ventilator’s settings, however, are strong determinants of when within the cycle speech can be produced. Speech produced with InPPV is often plagued by long pauses, short phrases, variable loudness, and variable voice quality.[30,33,34] These speech features are most easily explained by relating them to the tracheal pressure.

Figure 56-1



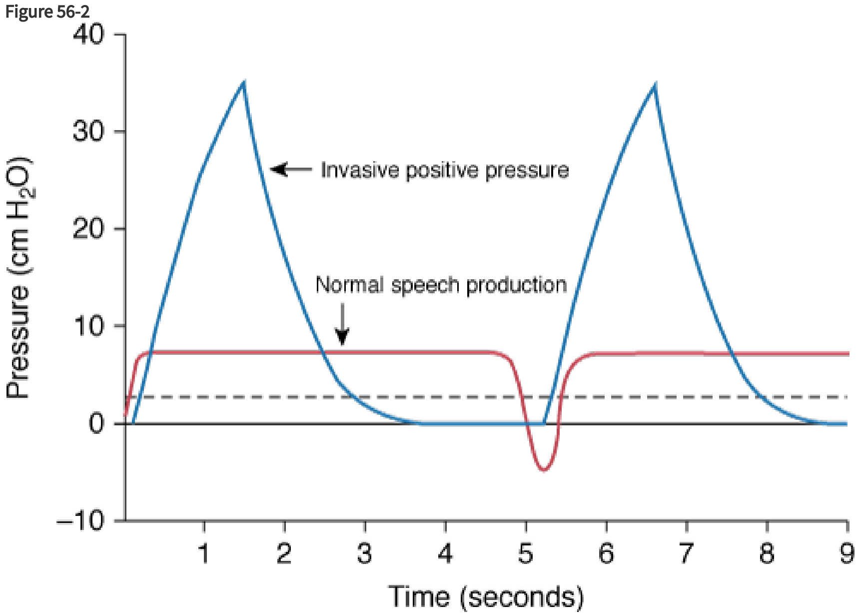
Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Inflated and deflated tracheostomy tube cuff during inspiration (*left*) and expiration (*right*). When the cuff is inflated, no flow reaches the larynx, and when the cuff is deflated, flow reaches the larynx during both inspiration and expiration. (Adapted and used, with permission, from Hixon and Hoit.[11])

Speech Production with Volume-Controlled Ventilation

Figure 56-2 illustrates tracheal pressure with volume-controlled ventilation compared with that of normal speech production. During normal speaking pressure is low and constant. This steady pressure allows vocal fold oscillations to be relatively constant in amplitude and waveform, translating perceptually to relatively constant loudness and voice quality. Also, pressure remains above the vocalizing threshold throughout expiration.

Figure 56-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Tracheal pressure during volume-controlled InPPV and during normal speech production. The *dashed line* shows the minimum pressure needed for vocalizing. (Used, with permission, from Hixon and Hoit.[11])

By contrast, during volume-controlled InPPV pressure rises quickly during inspiration, falls sharply when the expiration valve opens, and then remains below the vocalizing threshold for a substantial portion of the cycle. The rapidly changing pressure is the primary cause of loudness and voice quality problems, and the periods of low pressure (below the vocalizing threshold) explain the long pauses and the short phrases.

Ventilator Adjustments that Can Improve Speech

Abnormal magnitude and time course of tracheal pressure are at the root of the speech problems associated with volume-controlled InPPV. Therefore, the unifying principle for improving speech is to adjust the ventilator so as to generate a pressure that is better suited for speech production, while accommodating the patient's respiratory needs and comfort. Ventilator adjustments to improve speech are designed to (a) increase the portion of the cycle during which pressure remains above the vocalizing threshold (i.e., at least 2 cm H_2O), and (b) reduce the rate and magnitude of pressure changes during the period in which speech is produced. Many adjustments are possible, and the optimal combination for a given patient must be determined empirically. The success of this approach has been well documented in patients with spinal cord injury and neuromuscular disease,[30,34–36] and is likely to be effective in others. Examples of ventilator adjustments and their predicted influences on speech are given below.

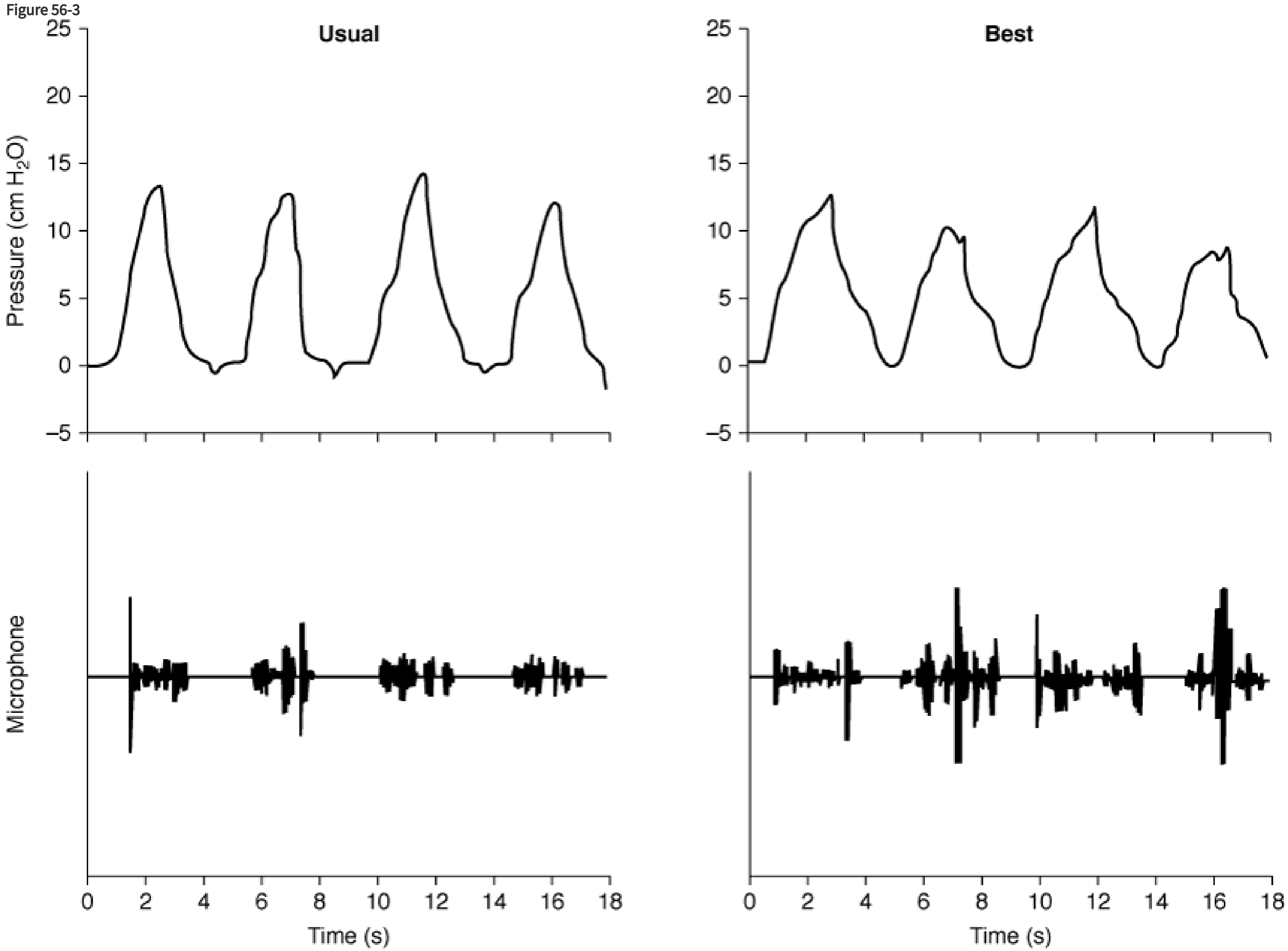
Lengthened Inspiratory Time, Positive End-Expiratory Pressure, and Reduced Tidal Volume

One of the easiest and most successful adjustment combinations is lengthened inspiratory time and the application of positive end-expiratory pressure (PEEP). When inspiratory time is lengthened, more speech can be produced during the inspiratory phase of the ventilator's cycle, because pressure is above the vocalizing threshold for a longer period. In general, as inspiratory time increases, the amount of speech produced during inspiration increases. Also, because the rate of pressure change is more gradual when inspiration is prolonged, loudness and voice quality variability may be reduced. The addition of PEEP extends speaking time during the expiratory phase by adding resistance to the ventilator's expiratory line, which keeps pressure above the vocalizing threshold longer. In most cases, pressure will eventually fall to zero secondary to flow through the larynx.

Sometimes it is physiologically appropriate to also reduce the ventilator-delivered tidal volume in patients whose tidal volumes are too large. When tidal volume is reduced, peak pressure is lowered, which helps to smooth out loudness and voice quality. Although patients are usually uncomfortable when tidal volume is reduced, they are often comfortable if PEEP is added first.

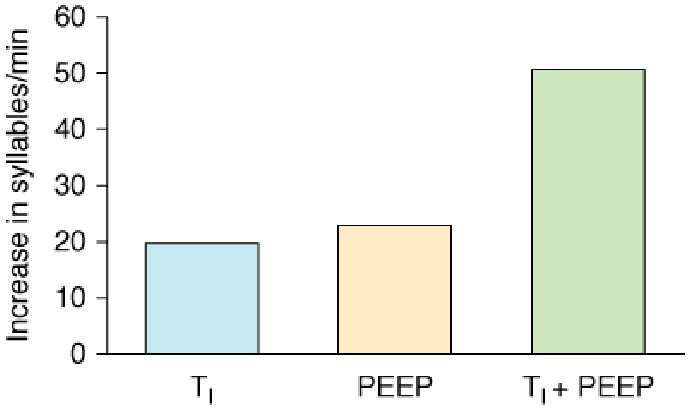
An example of the effect of these adjustments on tracheal pressure and the acoustic speech signal is shown in Figure 56-3 as a comparison of a patient's "usual" and "best" ventilator settings. Although the ventilator adjustments resulted in only subtle changes in the pressure waveform, the changes in speech were substantial and easily perceptible to listeners.[30] On average, an increase in inspiratory time of 17% to 25% combined with PEEP of 5 to 10 cm H_2O results in an increase in speaking time of 55% per cycle.[34] As shown in Figure 56-4, the increase in syllables/minute for lengthened inspiratory time and the PEEP adjustment is additive.

Figure 56-3



The tracheal pressure and speech signal under the usual (*left*) and best (*right*) ventilator conditions. The best settings represented the following adjustments: (a) inspiratory time was lengthened by 0.5 second; (b) 4 cm H_2O of PEEP was added; and (c) tidal volume was reduced by about 0.3 L. (Adapted and used, with permission, from Hoit and Banzett.[30])

Figure 56-4

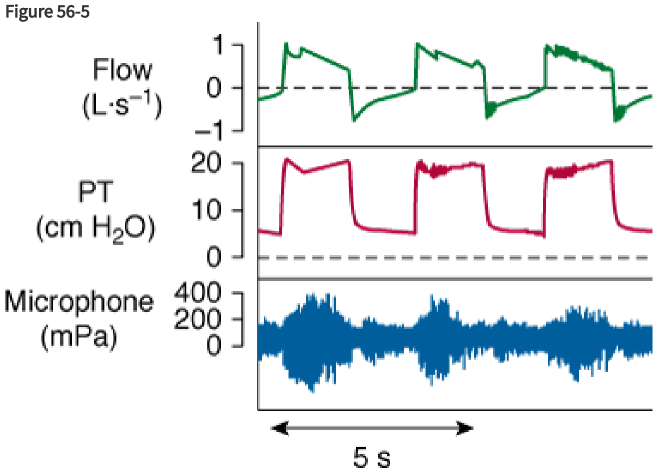


Changes (from usual) in syllables/minute for lengthened inspiratory time, PEEP, and combined lengthened inspiratory time (T_I) and positive end-expiratory pressure (PEEP). (Adapted and used, with permission, from Hoit et al.[34])

Pressure-Controlled Ventilation and Positive End-Expiratory Pressure

Pressure-controlled ventilation (and its variants, such as pressure-support ventilation) target a given airway pressure (rather than a volume). Thus, inspiration may be associated with a relatively steady pressure, somewhat similar to that associated with normal speech production (except higher). With this form of ventilation, pressure remains above the vocalizing threshold throughout inspiration. And, when pressure is steady, loudness and voice quality are more apt to be constant. When PEEP is added, pressure can be maintained above the vocalizing threshold during expiration. Figure 56-5 shows that vocalizing can continue throughout the entire ventilator cycle[35] (as it can with appropriate volume-controlled adjustments).

Figure 56-5

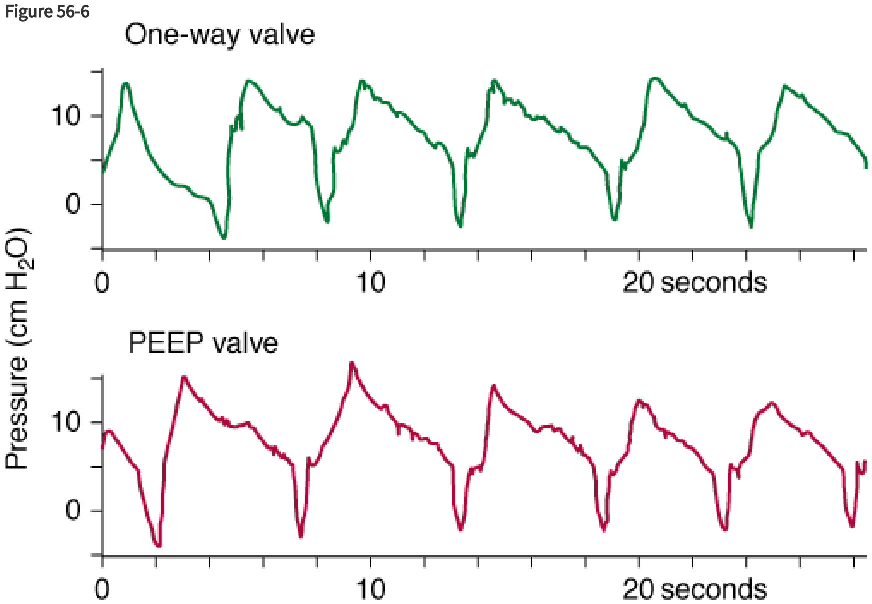


Flow, pressure, and speech signal during sustained vowel production with pressure-targeted ventilation and PEEP. (Reprinted with permission of the American Thoracic Society. Copyright © American Thoracic Society. Prigent H, Samuel C, Louis B, et al. Comparative effects of two ventilatory modes on speech in tracheostomized patients with neuromuscular disease. Am J Respir Crit Care Med. 2003;167:114–119. Offical Journal of the American Thoracic Society.)

Positive End-Expiratory Pressure Replaces a One-Way Valve

A one-way valve can improve speech in a patient with a tracheostomy who breathes spontaneously with a tracheostomy (see Chapter 40). Such a valve can also improve speech in a patient with InPPV; however, its use with InPPV can be dangerous. For instance, if the tracheostomy tube cuff is inflated and the valve is inadvertently left in place, the patient cannot expire. This can be lethal by causing asphyxia, by impeding venous return, or, if the pressure-limit device fails, by rupturing the lungs. A safer approach is to add PEEP.[37] With PEEP (Fig. 56-6), speech can be indistinguishable from that produced with a one-way valve,[34] even with PEEP as low as 5 cm H_2O.[38]

Figure 56-6



Tracheal pressure during speaking with a one-way valve and with positive end-expiratory pressure (PEEP) set to 15 cm H_2O. (Adapted and used, with permission, from Hoit et al.[34])

Risks and Benefits of Ventilator Adjustments

Ventilator adjustments can have adverse effects if not properly chosen and implemented. For instance, when inspiratory time is lengthened with volume-controlled ventilation, the patient may hypoventilate while speaking, because there is more time for inspiratory flow to "bleed off." If this happens, most patients experience air hunger and intuitively stop speaking to catch their breath,[39,40] but patients with diminished chemosensitivity may be at risk. If too much of the total cycle is devoted to inspiration, the shortened expiratory time may lead to dynamic hyperinflation, especially in patients with severe obstructive airways disease. Any adjustment that excessively increases intrathoracic pressure may reduce cardiac output and impede venous return. To minimize this risk, the lowest level of PEEP adjustment that reaps adequate speech improvement should be used, and dynamic hyperinflation (intrinsic PEEP) should be assessed following any changes to the ventilator. Also, ventilator adjustments can cause patient discomfort (e.g., "not enough air" or "too much pressure"). The potential for discomfort may be lessened if adjustments are made gradually and in small steps. Fortunately, most adjustments that improve speech also fortuitously improve comfort. Although there are risks to ventilator adjustments, they are usually minimal, especially when compared to the more serious risks posed by a one-way valve.

The most obvious benefits of these ventilator adjustments include improved speech, decreased speaking effort, and, in some cases, increased breathing comfort. There may be other benefits as well. For example, if a patient is chronically hyperventilated,[41] speech adjustments may move ventilation toward normal (e.g., because the patient is comfortable with a smaller tidal volume if PEEP is applied; see Clinical Scenario 2).

Clinical Scenario 2

JT, a man with a C2 spinal cord injury and InPPV, was referred to the speech-language pathologist for a speech evaluation. The evaluation revealed that JT spoke in short phrases (three to four syllables per breath) with excessively long pauses (up to 5 seconds), and he complained that he was "interrupted all the time." The speech-language pathologist recommended to JT's pulmonologist that ventilator adjustments be tested to improve his speech. The pulmonologist agreed and wrote the order, adding that "if the ventilator adjustments can also reduce his ventilation, that would be a bonus." Ventilator adjustments were tested (with the pulmonologist, respiratory therapist, and speech-language pathologist present) and it was found that lengthened inspiratory time, a PEEP adjustment, and reduced tidal volume improved his speech dramatically. Preadjustment and postadjustment measures revealed a moderate increase in arterial partial pressure of carbon dioxide (P_{CO_2}) (from 22 to 29 torr) and greater breathing comfort. The patient, his speech-language pathologist, his pulmonologist, and his family and friends were delighted with the outcome.

Behavioral Interventions that Can Improve Speech

Once appropriate ventilator adjustments have been made, behavioral interventions, guided by a speech-language pathologist, may further improve speech (see reference 11 for an extensive discussion of this topic). For example, linguistic strategies may be helpful for patients who continue to have long pauses between phrases despite ventilator adjustments. These linguistic strategies might take the form of pausing at appropriate linguistic junctures (e.g., at phrase, clause, or sentence boundaries) when speaking didactically, but intentionally pausing at inappropriate linguistic junctures (e.g., within a phrase or following a conjunction) when speaking conversationally to reduce the likelihood of being interrupted. It may be useful to teach buccal speech (i.e., "Donald Duck"-type speech) as a compensatory strategy. This type of speech, which is produced entirely within the upper airway and requires no drive from chest wall muscles,[42] can be used to add a syllable or two after the pressure has fallen below the vocalizing threshold.

Speech with Other Forms of Ventilator Support

Whereas speech can be produced during both inspiration and expiration with InPPV, it is produced only during expiration with other forms of ventilator support (see Table 56-1 and Chapters 16, 17, 18, and 62). Although research is sparse, ventilator adjustments and behavioral interventions have strong potential to improve speech produced with these other forms of ventilator support. Noninvasive positive-pressure ventilation (NPPV) and phrenic nerve pacer ventilation are discussed below as illustrations of how this might be done.

Table 56-1: Summary of Speech Produced Using Different Types of Ventilator Support

Type of Ventilator Support	Speech Production	Can Ventilator Adjustments Improve Speech?	Can Behavioral Interventions Improve Speech?
Invasive positive pressure	Inspiration/expiration	Yes	Yes
Noninvasive positive pressure	Expiration	Yes	Yes
Negative pressure	Expiration	Yes	Yes
Phrenic nerve pacer	Expiration	Yes	Yes
Rocking bed	Expiration	No	Yes
Abdominal pneumobelt	Expiration	Yes	Yes

Included here are the phase(s) of the ventilator cycle during which speech is produced and whether or not ventilator adjustments and behavioral interventions have potential to improve speech.

Noninvasive Positive-Pressure Ventilation

An important speech consideration when using NIPPV is selection of the patient-ventilator interface.[43] Choices for interfaces include mouthpieces, nasal pillows, nasal masks, and face masks. Of these, the clear choice for speech breathing is a mouthpiece that is not secured to the patient's mouth. To use this interface, the patient places his or her lips around the mouthpiece during inspiration and pulls the lips away during expiration. Thus, the patient's face is completely unencumbered during speech production. The next best interface options are nasal interfaces, because they allow the lips and jaw to move freely during speaking. Nevertheless, they encumber the nasal airway and can distort nasal consonants (*m, n,* and *ng,* in English). By far the least desirable interface is the face mask, because it encumbers both the oral and nasal airways, dampens and distorts the speech signal, and removes visual cues that a listener might use to help compensate for reduced intelligibility.

For many patients who use NIPPV, one of the best strategies for improving speech is to increase inspiratory volume.[11] In the case of the patient with neurologically complete chest wall paralysis, pressure (and loudness) at the beginning of expiration is determined primarily by inspiratory volume and respiratory system elastance, and the rate at which pressure drops (and loudness decreases) is determined by the rate at which volume is expended during speaking. By increasing inspiratory volume, speech can be louder (at least at the beginning of expiration) and last longer.

One way to increase inspiratory volume is to adjust the ventilator's tidal volume. Another way is to train the patient to use glossopharygneal breathing. Glossopharyngeal breathing can be effective, not only for increasing tidal volume, but also for replenishing volume (and pressure) while speaking.[44]

Phrenic Nerve Pacer Ventilation

In patients with paralysis from cervical spinal cord injury, speech with phrenic nerve pacer ventilation may be characterized by low loudness, fading loudness, short breath groups, and long inspiratory pauses.[45] Phrenic nerve pacers can be adjusted, within limits, to alter tidal volume, breathing frequency, and inspiratory time.[46] Such adjustments can increase speech loudness and duration and decrease the length of inspiratory pauses.

Tidal volume can be increased by applying an abdominal binder.[47,48] An abdominal binder or truss can improve speech in patients with chest wall paralysis who use phrenic nerve pacers[49] as well as in those who can breathe on their own.[50,51] Speech improvements include louder speech, longer breath groups, and better voice quality. Glossopharyngeal breathing is also a good strategy for increasing tidal volume in patients who use phrenic nerve pacers.

Which Patients Are Candidates for Speech Interventions?

Many of the speech interventions described in this chapter have been tested systematically in patients with neuromotor impairments who were chronically ventilated. Nevertheless, acutely ventilated patients, including those with other medical conditions, are also candidates for these interventions, especially cuff deflation. Cuff deflation and speech training are performed routinely in the postcritical respiratory care stepdown unit and even in the intensive care units at Massachusetts General Hospital.

There are several issues to consider when determining if a patient is a candidate for ventilator adjustments. Most importantly, ventilator adjustments must be deemed medically safe and should not be attempted if the pulmonologist, cardiologist, or other physician judges them to be unsafe for a given patient. The patient should also have motivation to speak, functional cognitive and language skills, and adequate laryngeal and upper airway control to benefit from such adjustments.

Behavioral interventions can be even more broadly applied than ventilator adjustments. There is a vast repertoire of behavioral interventions that include strategies for improving speech intelligibility, use of augmented communication, and many others.

Important Cautions

The initial deflation of the tracheostomy tube cuff should be done in a cautious and organized manner. To begin, the patient should be advised to expect flow of air through the mouth (as many are surprised by the sensation). Before deflating the cuff, the patient's oropharynx should be suctioned thoroughly and adequate oxygen saturation should be assured (often by increasing the fractional inspired oxygen concentration). An anesthesia bag supplied by oxygen should be attached to the tracheostomy tube and, at the moment of cuff deflation, a positive-pressure breath should be given to propel secretions that lie between the larynx and cuff into the pharynx for further suctioning. This technique minimizes aspiration of the retained secretions.

Cuff deflation usually causes patients to cough, likely from secretions above the cuff that move distally. If the cough does not subside quickly, it is useful to instill 5 mL of 2% lidocaine into the tracheostomy tube and occlude the opening of the tube briefly. The lidocaine will anesthetize the distal trachea and, when the patient coughs, the lidocaine will be propelled above the cuff as well. Patients who have had endotracheal intubation before tracheostomy are at increased risk for aspiration secondary to laryngeal dysfunction and should be evaluated accordingly before more than brief cuff deflation.

Cuff deflation may cause hypoventilation, hypoxemia, or dyspnea because some of the ventilator-delivered inspiration exits through the larynx. Thus, it is important to increase the tidal volume during cuff deflation. In addition, it may be beneficial to deflate the cuff only partially, rather than completely. The added resistance through the laryngeal pathway results in more of the ventilator-delivered tidal volume reaching the lungs.

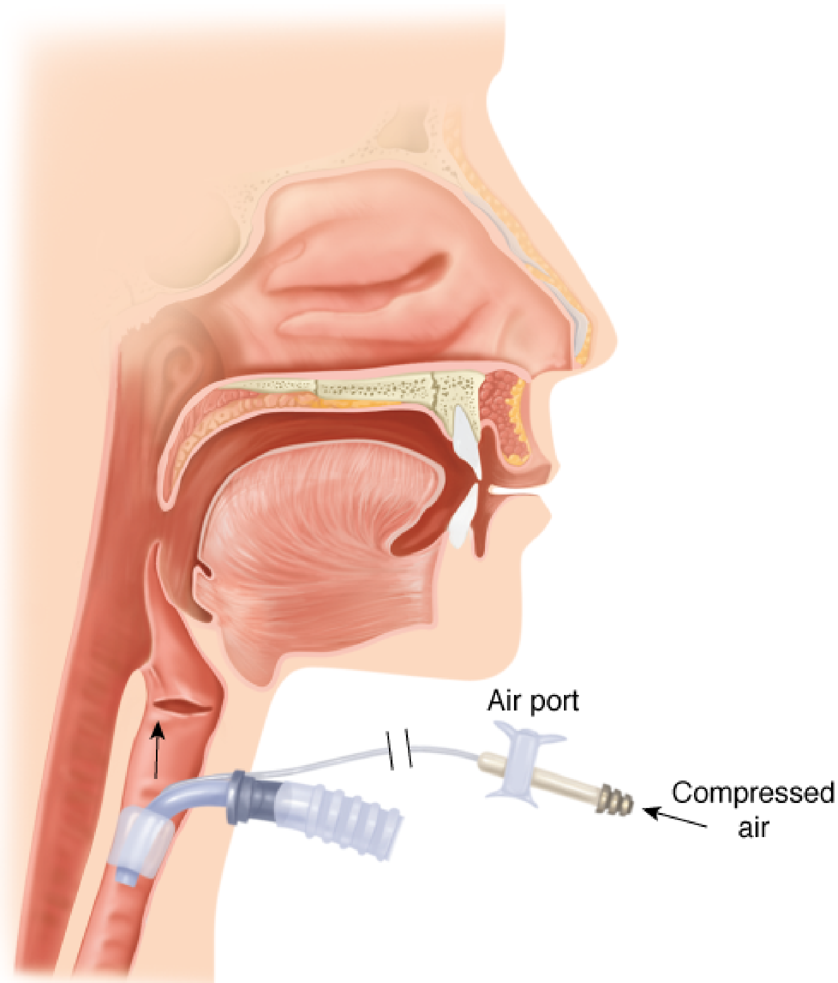
Although dynamic hyperinflation and intrinsic PEEP are observed rarely when inspiratory time is lengthened, it is nevertheless a risk in patients with obstructive lung disease. Positive end-expiratory intrathoracic pressure can lead to decreased cardiac output and hypotension. Cardiovascular reflexes usually compensate, but may be impaired in patients with autonomic dysfunction. These include patients with diabetes or with neurologically complete cervical spinal cord injury (particularly during the phase of spinal shock).

Ventilator adjustments to improve speech should only be made with the approval and oversight of a pulmonary or critical care physician. While adjustments are being made, cardiopulmonary variables should be monitored (e.g., O_2 saturation, heart rate, and blood pressure), and the patient should be systematically asked about comfort (see Chapter 55).

Alternatives to Ventilator-Supported Speech

Sometimes it is impossible for a patient to speak. When this results from severe laryngeal and/or upper airway impairment, it is usually necessary to provide the patient with augmentative and alternative communication (see Chapter 40). Some patients, however, with good laryngeal and upper airway function can benefit from a talking tracheostomy tube,[8,52,53] which provides a separate air source to produce speech while the tracheostomy tube cuff is inflated (Fig. 56-7). Flow is adjusted to determine what produces the best voice quality and is comfortable for the patient.[52,54,55] Common problems include discomfort from drying of the mucosa, crimping of the tubing, inability to vocalize, and inability to turn the air source on or off independently.[56]

Figure 56-7



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Talking tracheostomy tube. The air port is occluded to route flow from a compressed air source to the larynx for speaking. (Used, with permission, from Hixon and Hoit.[11])

Another device is called a voice tracheostomy tube.[57] The cuff on this specially constructed tracheostomy tube inflates during inspiration (preventing speech production) and deflates during expiration. Thus, nearly all of the ventilator-delivered inspiration reaches the patient's lungs, whereas some or all of the expired air can be used for speaking. A newer tracheostomy tube operates on similar principles and incorporates additional safety measures.[58] These devices may be useful for patients with poor laryngeal control, but most patients can speak using a conventional tracheostomy tube with a deflated cuff and some form of volume compensation.[59]

Roles of Health Care Professionals

Speech management for patients who are ventilator-supported requires a team of health care professionals. Pulmonary and critical care physicians and respiratory therapists operating under them can make substantial improvements in speech. Speech-language pathologists can further optimize speech by evaluating speech, language, and communication, and recommending a management plan to the medical team. If evaluation or management includes ventilator adjustments, it is important for a physician or suitable proxy to be present to assess cardiopulmonary effects of the adjustments as they are made. The physician in charge must decide whether such adjustments should be made permanently or only when essential for communication. Behavioral interventions (such as glossopharyngeal breathing and linguistic strategies) can be offered by the speech-language pathologist.

Important Unknowns and Future Research

Research on ventilator-supported speech has largely been limited to patients with spinal cord injury and neuromuscular disease. Much needs to be learned about the influence of ventilator adjustments on speech in other patients, especially those with obstructive airways disease. The speech effects of ventilator adjustments with certain forms of ventilator support (such as NIPPV) and various behavioral interventions also have yet to be studied systematically.

Summary and Conclusions

For most people, speech is an important requisite to good quality of life. Thus, a patient should be given the opportunity to speak, if medically possible. For the patient who is ventilated invasively, this means deflating the tracheostomy tube cuff. For the patient who is already able to speak, there are often ways to improve speech by modifying the tracheal pressure. Ventilation requirements always have primacy over speech improvements, but these are often complementary, rather than competing, goals. A patient's speech can be optimized through the collaborative efforts of competent and creative physicians, respiratory therapists, and speech-language pathologists.

Acknowledgments

This chapter is dedicated to the late Susan Kropoff, who used a ventilator for nearly 40 years, and who has taught us more about ventilator-supported speech than anyone else we know. We are grateful to Thomas J. Hixon and Robert W. Lansing for their helpful suggestions on this chapter.

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